



FOUNDATION FOR WATER

help water support life

PLANT RESEARCH

Plant Growth Processes Influenced by Water
Treated with Rhythmic Dynamic Flow

Summer Autumn 2015





Overview of the Research Project

The main question was if Flowform rhythmical, dynamic flow treatment of irrigation water caused plants to express their archetypal 'morphological' type more in their forms and quality character more than in control plants not irrigated by such water.

Our purpose was to examine this in general terms with specific trials to see if indeed it would be valuable to run more in-depth trials to levels of statistical significance in a following growth season.

A second question asked how timing of germination was an important factor, and on the basis of years of successful Biodynamic calendar usefulness , this was a question that asked if Flowform water treatment intensified the etheric formative forces at work in plant development in relation to the four ethers and plant types which the BD calendar is designed to help practitioners with.

A third aspect involved inverting this second question, to see if Flowform water increased the effects of unhelpful planting times. This could help show that the cosmic information at work in life forms energetic processes at any time would be increased by Flowform water.

The assumption behind this research is that water, including all water-based fluids, is the medium within organic life forms that cosmic formative information from stars and planets engages with to influence energetic growth processes active in nature.

Within the anthroposophic world conception this can be understood to mean that such water might increase the capacity of the archetypal beings of an organic life form to incarnate more deeply and expressively into its living form, via the spiritual light-based information sent into it via water for nature spirits to work more effectively with.

Description of the Project

Four plants expressing the character of the four ethers were chosen to grow and observe, in two equal sets of Flowform and Control groups. Biodynamic seeds of each plant type were planted and irrigated in the first set of trials *once only* at their planting 'moment', half of them with Flowform treated water and half with Control water, which was treated through an oxygenating waterfall. (refer photo)

In the second set of plant growth trials started late in the summer, (that are not yet complete), the plants were watered throughout their growth separately with Flowform and Control waters.

Most of the information following concerns plants from the first set of trials watered only once by Flowform and treated Control water at their time of planting. The plants were irrigated with non-treated spring water for the rest of the growing season in the first set of trials .

All other aspects of the experiments, such as pumps, soil, containers, sources and amounts of the spring and rain water, timing of watering etcetera were the same as much as possible. All plants were grown in a greenhouse environment, with no rain or dew affecting them.



Capsicum – Warmth Ether: expressed as a fruit plant type, planted at a fruit time.

Description: If the hypothesis was accurate then Flowform influenced water given to sweet green pepper would give rise to more fruit with a better quality.

10 plants were planted direct into large pots, alternating side by side. The Flowform influenced plants grew faster, had 35% more branches, and more flowers on average.



Control plants fruited earlier but had fewer and smaller fruit on average. These two plants indicate the pattern of the set of ten.

Capsicum – Warmth Ether: fruit plant type, planted at a fruit time.

Photo shows the alternating plants irrigated from left by Control water and then next by Flowform water. This continues c,f,c,f,c to the right. One of the Control plants off the picture to the right appeared more like the Flowform plants but it didn't greatly change the averages.



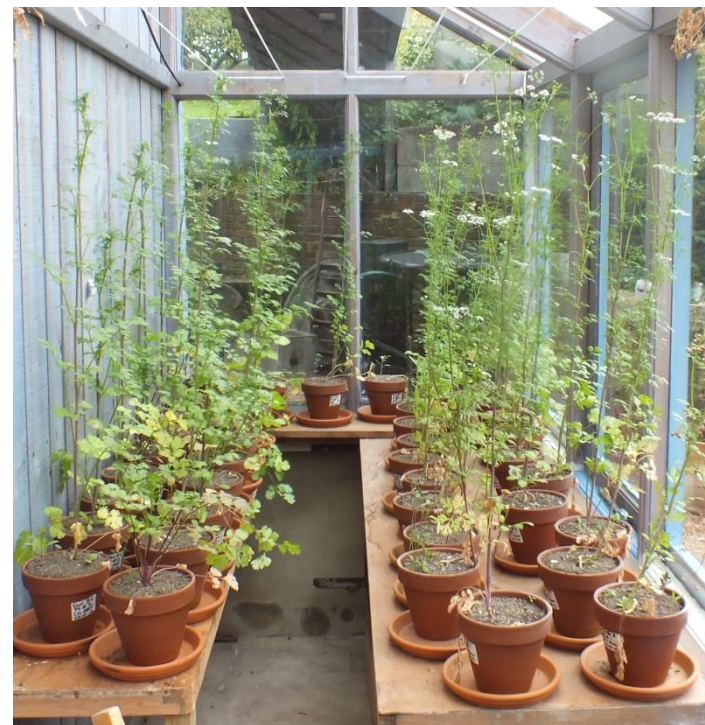
Coriander – Light Ether: flower plant type, planted at a flower time.

Description: The influence of the light ether should reduce leaf size, and make it more fragmented, or have less leafy mass, as if 'eaten into' by light. This is how plants are affected when growing in strong sunlight throughout a growing season. 20 plants each of Flowform and Control were irrigated in root trainers and later transplanted into larger root boxes, with glass fronts.

The young plants showed the influence of extra light from an early stage, as indicated in the photo on the left below, an approximate average of the whole set of plants.

For adult expression of untransplanted root trainer plants please see over next page.

Plants to the right here, are two sets of coriander. To the left are Flowform and Control coriander planted at a 'leaf time' and to the right Flowform and Control coriander planted at a 'flower time'. They show considerable difference. Note: the greenhouse has abundant light from above.



Coriander – Light Ether: flower plant type, planted at a flower time.

Average samples showing smaller, leaf reduced, earlier expression coriander as if influenced by more light, yet they grew side by side.



Lettuce – Chemical Ether: leaf plant type, planted at a leaf time.

Description: The expectation would be that the leaves would be more expressive morphologically and the sizes would be larger and more fleshy, from Flowform treated water compared to Control Water irrigated plants. This is the direction of the results, with fresh and dry weights of all the lettuces together indicating this. The shapes of the plants also indicated this (see over). As a reminder, these were watered only once at time of planting.

We did blind taste testing with lettuces and radishes, and found that overall Flowform influenced plants tasted sweeter, were softer to chew and had a more subtle, long lasting flavour.



Lettuce – Chemical Ether: leaf plant type, planted at a leaf time.

Description: 30 lettuce plants were grown, with these photos indicating a visual average of their growth.

Roots tended to be larger and held onto the earth when being cleaned for weighing. (Below)

The plants more and larger leaves as seen (right) where average Flowform and Control plants are compared, with the first leaves at the base.



Radish – Life Ether: root plant type, planted at a root time.

Description: Roots were conjectured to become larger with Flowform water. The total 50 radish Responded strongly to this treatment, with better germination strike rates, more weight and size in the roots and also clear change in taste testing towards subtle hot flavours, combined with sweetness and smooth texture.

The plants also had larger leaves on average and also avoided being eaten by pests more. (See over)



Radish – Life Ether: root plant type, planted at a root time.

Indications of larger and less eaten leaves of typical plants from the set



Separately, we can note that two sets of 'unhelpful time' plantings, masked any advantages that Flowform was showing in 'helpful time' plantings, and that the taste testing gave strong indications of coarser texture and a bitter, more hot taste in radishes

Radish – Life Ether: root plant type, planted at a root time.

Description: Leaf destruction by caterpillars on Control radishes, growing beside Flowform plants. See over for a front view.



Radish – Life Ether: root plant type, planted at a root time.

View of the same plants from the previous page but from the front



Radish – Life Ether: root plant type, planted at a root time.

Set 2, x10 radishes each type, watered throughout growing period with Flowform and Control water. They are still growing and so only visual is possible, but Flowform ones are considerably larger.



Overall Summary

The aim of the experiment was to see if the hypothesis that rhythm combined with dynamic flow – made up of laminar, harmonic and turbulent movements – would increase the ‘type expression’ of very different plants. The overall result – at this early stage of assessment – supports this outcome.

This would indicate that Flowform water may hold different ‘life supporting information’ in it, different to the otherwise same Control water, in such a way that Nature can carry out its activities more in tune with plant – and possibly other – organic processes

While, much of the data collected is not yet analysed due to growth trial timing, we are reassured that further trials during 2016 which could allow for statistical analysis, and more methods of assessment including spiritual scientific observation would be a valuable use of resources. We would also plan to have plantings in our field trial garden, while also increasing our greenhouse space.

As such this is a preliminary report, based on visuals and measurements and can be followed up further on the basis of further funding to continue the overall project.

23rd October, 2015

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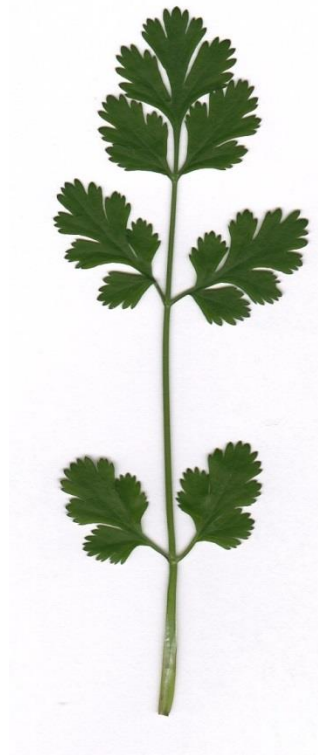
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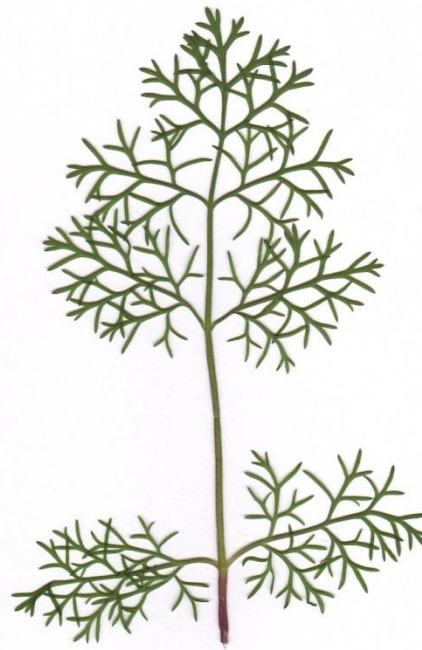
Radish taste testing

The following pages outline coriander research in 2013 compared to earlier lettuce research which led to the 2015 research reported on in the previous pages

Water influenced by rhythmical dynamic flow brought more light via plant roots to influence plant morphology as if they were growing in bright sunlight



Usual young early-stage Coriander leaf growing in moderate sunlight with water uninfluenced by rhythm



Same stage Coriander leaf growing in the same moderate sunlight with same soil but with water influenced by rhythm and dynamic flow

This strongly serrated leaf is typical of strong sunlight influenced growth. The extra light can only have affected the morphology by entering the roots via water

The leaves shown previously were taken from Coriander plants like these experimental samples. Over 80 plants were grown in the same environments with (the same) water which was influenced by Flowform and non-Flowform water and turned over with the same pumps. (Left – Flowform water. Right – non-Flowform)



Typical growth as if in strong sunlight – yet the sunlight was moderate strength

Typical growth as if in moderate sunlight – which was indeed the actual environment

Foundation for Water wants to recreate morphology trials with varied plant types to test the hypothesis that rhythmic dynamic flow brings cosmic formative forces more deeply into its growth process *relative to the plant type's etheric emphasis*.

Below it can be seen that lettuce behaved in the opposite way to coriander, becoming more leafy under Flowform influence – which is what lettuce is. This implies a strong chemical ether at work in the lettuce.

Coriander however is a light influenced plant as can be seen by its leaves and starry flowers. It had a stronger light ether at work as evidenced by its morphology response.

Work with carrots has shown in the past that the roots become bigger but not the leaves.



Lettuce leaf morphology trials – most Flowform influenced left, and control, right